Data Mining

Block 1 - What it is, and how not to fool yourself

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About me

- Data Science manager & data scientist - specialising in credit risk and fraud detection
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- MSc in Computer Science and in Automatic Control & Robotics, Warsaw University of Technology
- Homepage: `datadeer.pl`

Expect many examples from retail credit risk - not because it is the course domain, but because it is the field closest to me.

Today, hour by hour

1. **Course logistics** - rules, dates, grading ~10 min
2. **What data mining is** - the map of learning problems, CRISP-DM ~35 min
3. **EDA** - how to actually read a dataset ~50 min
4. **Missingness & feature engineering** - today's core skills ~35 min
5. **The craft** - habits that save beginner data scientists ~20 min
6. **Lab** - notebook `01_eda_missingness` ~30 min

Breaks roughly every 50 minutes. Questions welcome at any point.

The deal, in one slide

Seven Tuesdays, 17:10–20:30

- 6 teaching blocks + 1 for defences & exam
- Language: English; code in Python
- Running examples: retail credit risk

Passing

- Project **50%** + Exam (term 0) **50%**
- Pass with $\geq 60\%$ overall
- Project defended in Block 7; exam same day

Where we are going

#	Topic	Date
1	Data mining, CRISP-DM, EDA & **missingness**	06.10
2	Linear & logistic regression	13.10
3	Trees, ensembles (RF, GBDT), honest evaluation	20.10
4	Clustering & dimensionality reduction	27.10
5	Pattern discovery (Market Basket) & text mining	03.11
6	Neural networks & responsible ML	10.11
7	Project defences + exam (term 0)	17.11

One running example throughout: modelling a credit portfolio well.

The project: what exactly you will build

- In a team of 3: a full mini CRISP-DM cycle on a dataset of your choice - business framing → data preparation → **at least two model families** → correct evaluation → interpretation → a recommendation.
- Data: any real dataset that can carry the cycle - a meaningful business decision, a predictable target, enough rows for an honest split (Kaggle, UCI, open data…). Declared and approved via the topic description; unsure? ask early.
- The project **ends with a presentation of results** in Block 7 (the last class), ∼6–8 minutes + questions.

Deliverables - both uploaded before Block 7 starts

(i) all code, as a reproducible notebook; (ii) the exact presentation you will show. What you upload is what you present.

The project: timeline & deadlines

Deadline	What is due
before Block 2 (13.10)	teams of 3 formed and registered
before Block 3 (20.10)	topic chosen + a short written description (the decision, the dataset, the target, the planned two model families)
before Block 7 (17.11)	**upload all code + the final presentation**
Block 7 (17.11)	presentation of results + Q&A

No topic idea? Consult before the deadline - e-mail or after class. That is what the consultations are for.

The project: a sensible week-by-week split

Week	Focus - with the suggested methods
1	find two teammates; first contact with a candidate dataset
2	topic + description; EDA: distributions, target, missingness
3	data prep (impute + indicators, one-hot); **baseline:** logistic regression in a pipeline, honest split; leakage audit
4	second family: random forest / GBDT; evaluation kit: CV, ROC/PR, calibration, cost threshold
5	optional: clustering + PCA, rules, or TF-IDF; interpretation → recommendation; presentation skeleton
6	freeze notebook (Restart & Run All); slides; rehearse; upload
7	present the results

Each week's block teaches exactly what that week's project step needs - stay on this schedule and the project builds itself alongside the course.

The project: grading & the AI rule

Rubric (50% of the course)

- framing & problem definition - 10%
- data prep & missingness - 15%
- modelling: $\geq$2 families - 15%
- evaluation, leakage-aware - 20%
- interpretation & recommendation - 15%
- reproducibility - 5%
- presentation & defence - 20%

Use of AI

- allowed - but you are responsible for every line you submit
- the more the project feels produced *solely* by AI, the harder the defence questions
- expect to explain and modify your code *live*

Note the weights: the presentation is a fifth of the grade - a correct analysis you cannot communicate and defend is worth less than you think. Raw accuracy alone buys almost nothing.

Prerequisites & tooling

What we will work with

- **Python** - all labs and the project are in Python
- **Jupyter notebooks** - run locally or on Google Colab
- Worth refreshing before next week: `pandas` (DataFrames, filtering, groupby, joins)

Setting up your environment

- Recommended: `uv` - one tool for Python versions, virtual environments, and packages
- `uv sync` in the course repo gives you a ready-to-run environment

Rusty?

Notebook `00_python_primer` is your self-paced warm-up.

What data mining *is* - and is *not*

It is

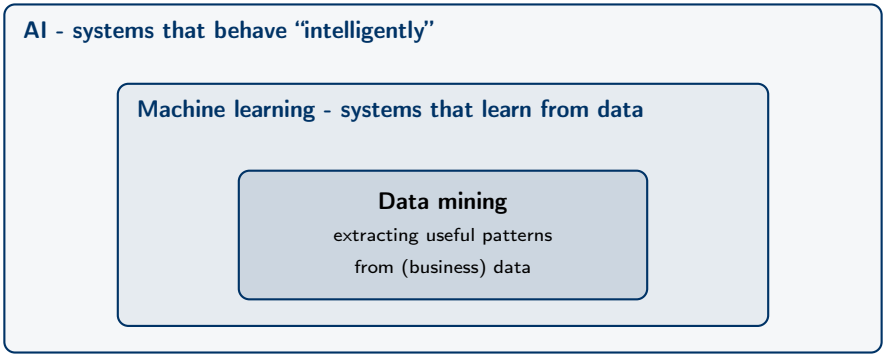
- finding useful structure in data
- to support a *decision*
- with methods that are honest about uncertainty

It is not

- running every algorithm and keeping the highest accuracy
- "the model said so"
- a substitute for understanding the business problem

We start from the decision, then choose the method - never the reverse.

Data mining, statistics, ML, AI - who owns what?



- **Statistics** underpins all three rings: inference, uncertainty, experimental design.
- In practice the borders are soft - employers say "data science" and mean the union.

Three ways a machine can learn

Supervised

- data comes with labels
- learn to predict the label for new cases
- *will this customer default?*

Unsupervised

- no labels - find structure
- clusters, patterns, compressed representations
- *which customers are similar?*

Reinforcement

- an agent acts, gets rewards, adapts
- *which collections action to try next?*

This course: **supervised** (Blocks 2–3, 6) and **unsupervised** (Blocks 4–5). Reinforcement learning we only mention - so you know where it sits on the map.

Supervised learning comes in flavours

Task	Target	Credit example
Binary classification	two classes	`default` $\in \{0, 1\}$
Multiclass classification	several classes	complaint category; rating grade
Regression	a number	loss amount; recovery rate
. . . and more	ordinal, ranking, survival time	months until default

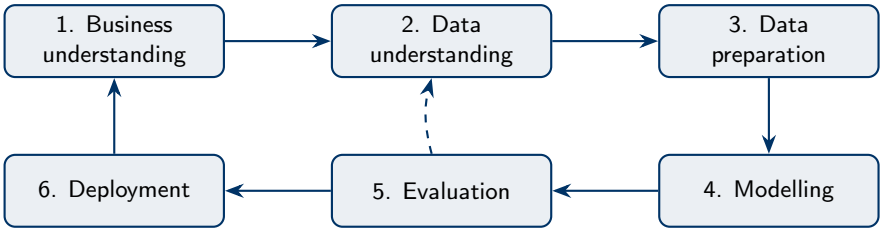
Our focus: binary classification and regression.
The rest we will meet in passing - enough to recognise them in the wild.

From business question to learning task

Business question	Learning task	Covered in
Will this applicant default?	binary classification	Blocks 2–3, 6
How large will the loss be if they do?	regression	Blocks 2–3
Which complaint team should read this email?	multiclass classification	Block 5 (mention)
Do our customers form natural segments?	clustering	Block 4
Which products are bought together?	association rules	Block 5

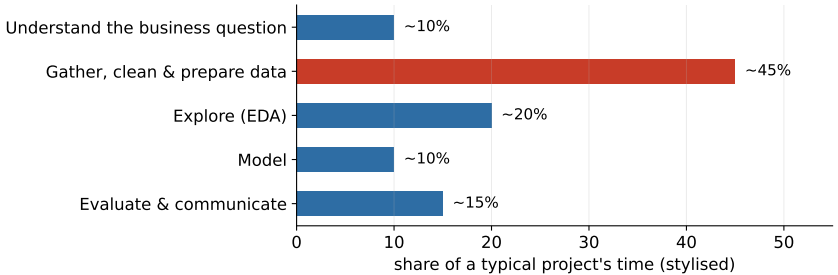
The translation step is your job - no algorithm does it for you.
Getting it wrong here invalidates everything downstream.

The spine of the course: CRISP-DM



Today lives in boxes **1 and 2**. The dashed arrow is the point: evaluation sends you back to understanding.

Where a project's time actually goes



- Modelling - the part courses spend most time on - is a small slice of real work.
- That is why this course starts with data understanding, not with algorithms.

Our business problem for the whole term

The lender's question

Given what we know about a customer at application time, how likely are they to default? And what should we *do* about it?

- Each row = one credit customer; target `default` $\in \{0, 1\}$.
- Realistic ingredients: income, debt-to-income, utilisation, prior delinquency, existing loans, savings, purpose...
- Realistic problems: class imbalance, missing data, and leakage - all of which we meet today or next week.

Today: the synthetic portfolio via `finance_data.load_credit()`; from Block 2, the real Taiwan portfolio via `load_taiwan()` - same lender's question.

First decision nobody teaches: what is a *row*?

The unit of analysis - what one row of your table represents:

- one **application**? one **customer**? one **customer-month**? one **transaction**?

Why it matters

- The unit fixes what the prediction *means* - and when it can be used.
- A customer with 3 loans: three rows or one? Your metrics change either way.
- Mixing units (some rows customers, some applications) silently double-counts people.

In our portfolio: one row = one customer with one loan application.

Meet the dataset

Column (sample)	Type	Notes
`customer_id`	identifier	never a feature!
`age`, `monthly_income_pln`	numeric	income has missing values
`housing`, `purpose`, `region`	categorical	no natural order
`checking_status`	categorical (ordinal)	none $<$ low $<$ medium $<$ high
`credit_utilization`, `debt_to_income`	numeric ratios	key risk drivers
`months_since_last_delinquency`	numeric	structurally missing
`collections_contact`	0/1	a trap - more in Block 3
`default`	0/1	the target

4,000 customers, 18 columns; German Credit (OpenML) or a synthetic Polish portfolio - same interface.

The data you will never see: selection bias

Ask of every dataset: how did rows get *into* this table?

A lender's portfolio contains only approved loans. Rejected applicants have no `default` outcome - they are simply absent.

- Our model learns on the *survivors of yesterday's policy* - a form of survivorship bias.
- For applicants unlike anyone previously approved, the model extrapolates - its confidence there is borrowed, not earned.
- Credit scoring has a name and a toolbox for this (*reject inference*); most other domains quietly ignore it.
- Same pattern everywhere: churn models see only acquired customers, hiring models see only past hires.

Selection bias cannot be fixed by more data from the same source.

Before any model: look at the data

Data understanding checklist

1. Shape, columns, data types.
2. Target balance - how (im)balanced is `default`?
3. Univariate: distributions, implausible values, units.
4. Bivariate: which features move with the target?
5. Data quality: duplicates, outliers, and *missingness*.

Skipping this step is the most common way a data-mining project quietly fails.

First contact with a DataFrame

Four commands before anything else:

```
df.shape                    # how much data do I have?
df.dtypes                   # what does pandas think each column is?
df.head(10)                 # do the values look like the docs promised?
df.describe(include="all")  # ranges, counts, uniques, NaNs
```

What you are looking for

- numbers stored as strings (`"12 000"`, `"N/A"`, `"-"`)
- categorical levels that differ only by case or spelling
- minima / maxima that make no physical sense
- a `describe()` count smaller than `len(df)` - that is missingness

Duplicates and keys

Two questions for every table

- What *should* uniquely identify a row? (the key - here: `customer_id`)
- Does it? `df["customer_id"].is_unique`

Where duplicates come from

- repeated exports glued together; joins that multiplied rows; customers applying twice

Why it is dangerous

- duplicated rows are double-counted evidence - metrics look better/worse than reality
- a duplicate that lands in both train and test set inflates every score (a leakage cousin - Block 3)

Know your variable types

Type	Example	Watch out for
Numeric, continuous	income, DTI	skew, outliers, units
Numeric, discrete	# existing loans	often better treated as categories
Categorical, nominal	purpose, region	rare levels, typos, encodings
Categorical, ordinal	checking status	order matters - keep it
Dates & times	application date	time zones; enables leakage
Text	complaint narrative	needs its own pipeline (Block 5)
Identifiers	`customer_id`	never a feature

The type decides the plot, the summary statistic, *and* the encoding.

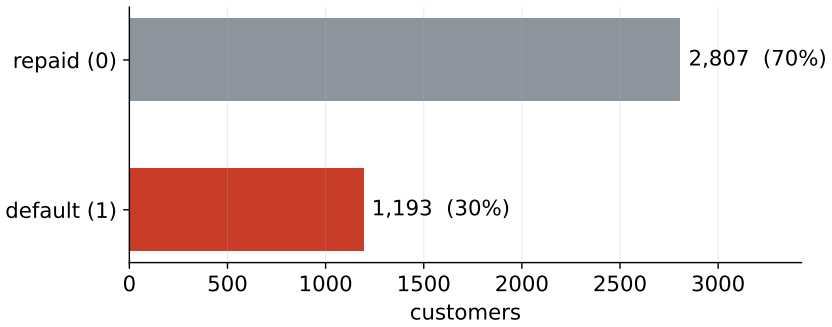
Scales of measurement (Stevens)

Scale	Defining property	Meaningful ops	Example
Nominal	categories, no order	$=, \neq$	purpose, region
Ordinal	ordered, gaps unknown	$<, >$	checking status
Interval	equal gaps, no true zero	$+, -$	temperature in °C; dates
Ratio	true zero exists	$\times, \div$	income, loan amount, age

Why it matters

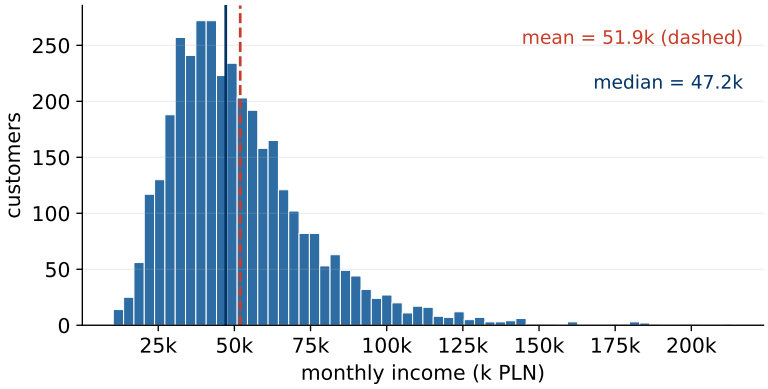
- The scale limits the statistics: a mean of nominal codes is nonsense; medians need at least ordinal.
- "Twice as much" needs a ratio scale: 40°C is not twice as warm as 20°C, but a 40k loan *is* twice a 20k loan.
- Encoding an ordinal variable as one-hot throws its order away; encoding a nominal one as integers invents an order that isn't there.

The target comes first: class balance



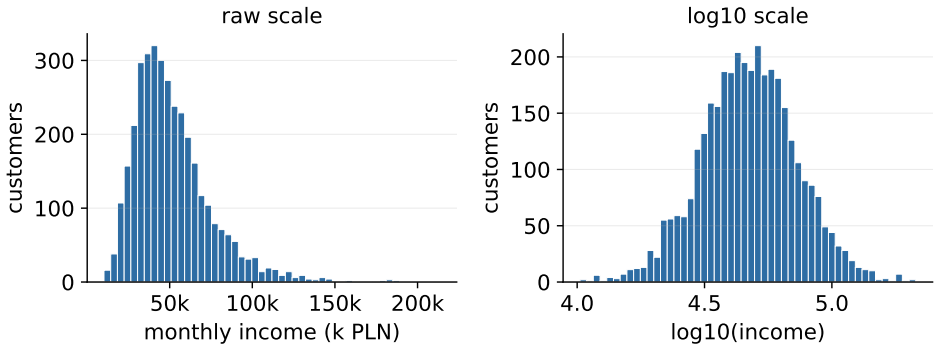
- A model that always says "repaid" is right 7 times out of 10 - and completely useless.
- Accuracy is therefore *not* the metric for imbalanced problems (proper treatment in Block 3).
- Real PD portfolios are far more extreme: 1–5% default rates.

Univariate EDA: the shape of a distribution



- Income is right-skewed: mean > median - a few high earners pull the mean up.
- Skew is information: it tells you which summary to report and whether a transform will help.

Heavy tails? Change the lens



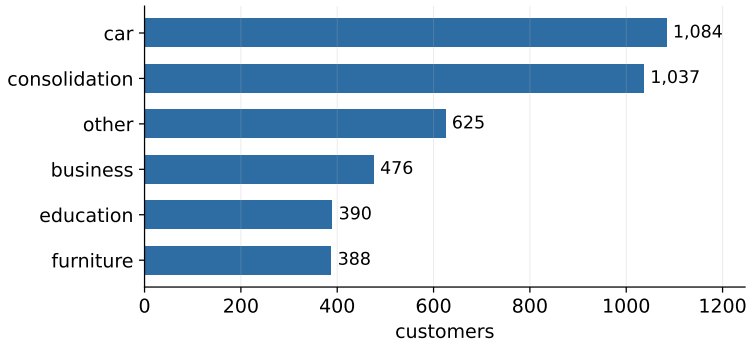
- On a log scale the same variable becomes nearly symmetric.
- Monetary amounts (income, savings, loss) almost always deserve a log look - multiplicative thinking fits money.

Sanity checks: implausible values

Red flag	Example	Likely cause
Impossible values	age $= 150$; income < 0	entry error, sentinel codes
Sentinel numbers	income $= 999999$	"missing" coded as a number
Unit mixes	salary 3,000 vs 36,000	monthly vs annual, PLN vs EUR
Too-round values	everyone earns exactly 5,000	defaults filled by a form
Frozen columns	one value everywhere	broken export, dead field

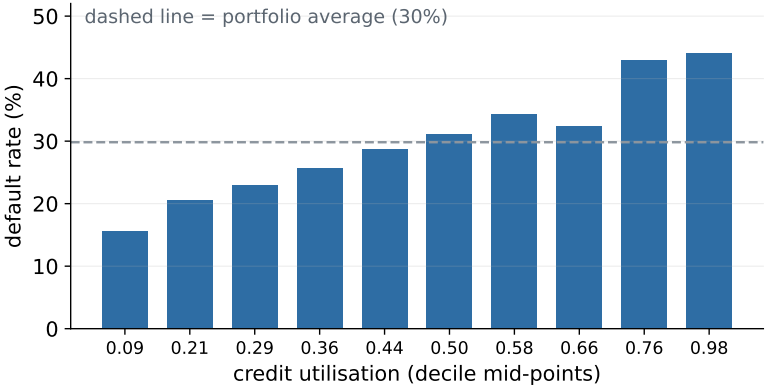
Every red flag is a question for the data owner, not a value to silently "fix".

Categoricals: levels and rare categories



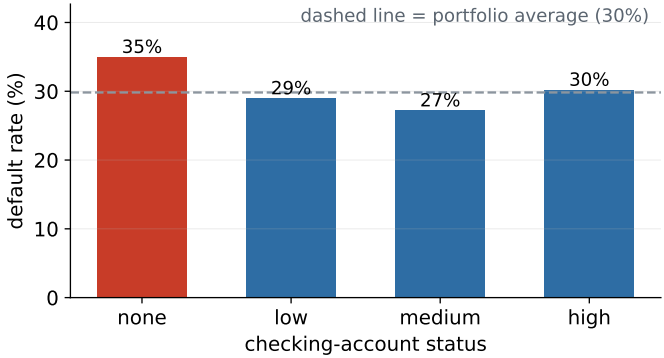
- Check levels: typos and case variants create phantom categories.
- Rare levels (<1–2% of rows) often get grouped into `other` - but only *after* checking they are not special (e.g. a tiny, very risky segment).

Bivariate EDA: numeric feature vs target



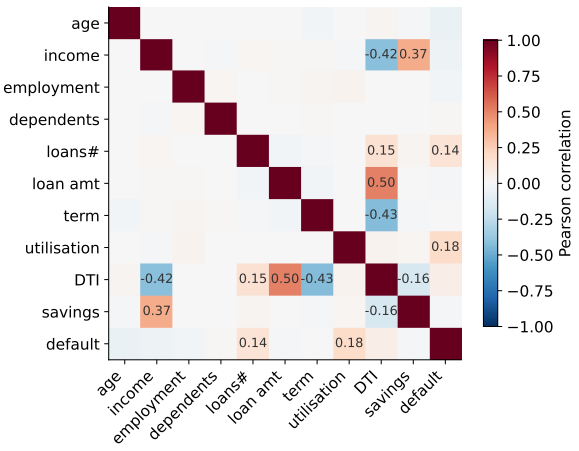
- Bin the feature, plot the default rate per bin - the shape of the risk relationship appears.
- Monotone and steep $\Rightarrow$ a strong candidate feature (utilisation is a classic credit driver).

Bivariate EDA: categorical feature vs target



- "No checking account" is the riskiest group - the bank cannot see those customers' cash flows.
- Compare each level to the *portfolio average*, and mind small groups: a rate computed on 30 customers is noise.

A first map: the correlation matrix



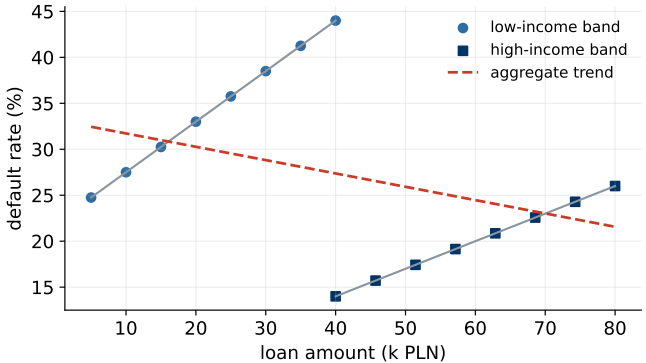
- One picture: which features move together, and which move with the target.
- Utilisation and the number of existing loans stand out for `default` - current debt load beats income here.
- Strongly correlated feature pairs ($|r| > 0.8$) are candidates for dropping one.

Correlation: handle with care

- **Correlation is not causation.** Ice cream sales correlate with drownings; summer causes both.
- **Pearson sees only straight lines.** A perfect U-shaped relationship can have $r \approx 0$.
- **Outliers can manufacture correlation.** One extreme point can turn $r = 0.1$ into $r = 0.7$.
- **Rank alternatives exist.** Spearman correlation asks only "do they move in the same direction?" - more robust for skewed data.

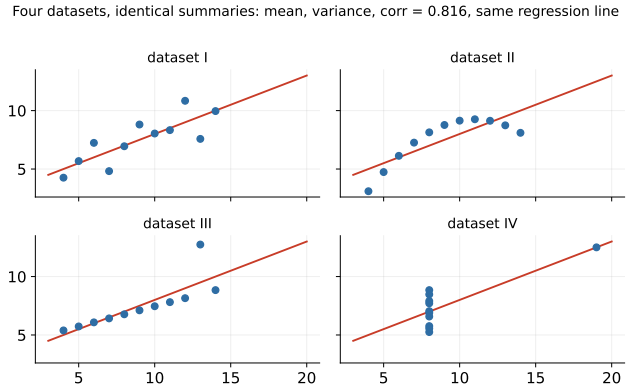
A correlation matrix generates hypotheses, not conclusions.

Simpson's paradox: the aggregate can lie



- In aggregate, bigger loans look *safer* - because high-income customers take the big loans.
- Within *every* income band, bigger loans are riskier. Both statements are true; only one is useful.
- Before concluding from any bivariate plot, ask: compared within what? (Illustrative data.)

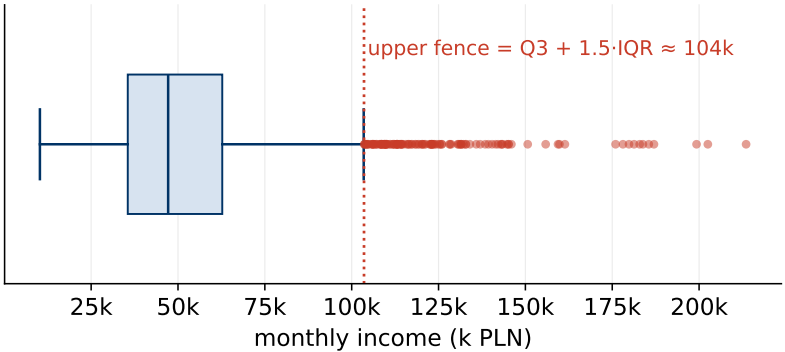
Why you must plot: Anscombe's quartet



Same mean, same variance, same correlation, same regression line - four completely different stories.

Summary statistics hide; plots reveal.

Outliers: how to spot them



- The boxplot's **IQR rule**: flag points beyond $Q3 + 1.5 \cdot \text{IQR}$ (or below $Q1 - 1.5 \cdot \text{IQR}$).
- A convention, not a law - for skewed variables it flags many perfectly genuine values (log first, then look again).

Outliers: what to actually do

1. **Investigate first.** Error or real? A 2 mln PLN income might be a typo - or a real customer who matters.
2. **If it is an error**: fix from source, or set to missing (do not invent a value).
3. **If it is genuine**: keep it; consider capping / winsorising or a log transform so it stops dominating.
4. **Document the rule** - "we capped income at the 99th percentile" must be written down and applied identically to new data.

Fraud-detection perspective

In fraud, outliers are not noise - they are often *the signal*. Whether an outlier is a problem depends on the business question.

EDA is a loop, not a phase

The rhythm

1. ask a question
2. make the one plot that answers it
3. the answer raises two new questions
4. repeat until surprises stop

Keep an evidence log

- every finding: one sentence + the plot
- "income is MNAR-missing for high earners" beats 40 unlabelled histograms
- your future self (and your team) will re-read it

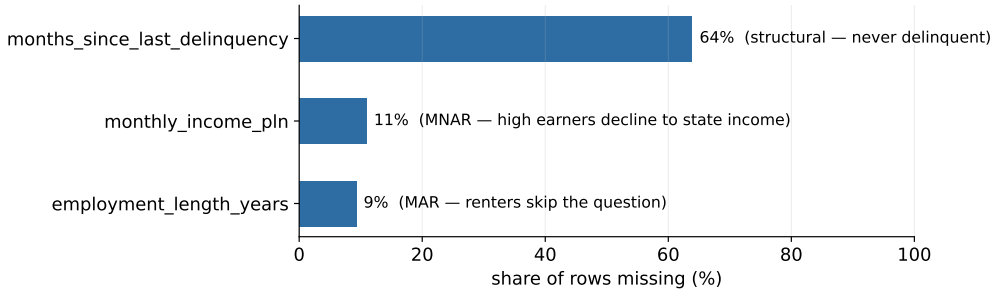
EDA never really ends - evaluation (CRISP-DM's dashed arrow) sends you back here.

Missing data is not one thing

Type	Meaning	Credit example
MCAR	missing completely at random	a system glitch drops a field
MAR	depends on *observed* data	renters skip employment length
MNAR	depends on the *unobserved* value	high earners hide income
Structural	the value cannot exist	"months since last delinquency" when there was none

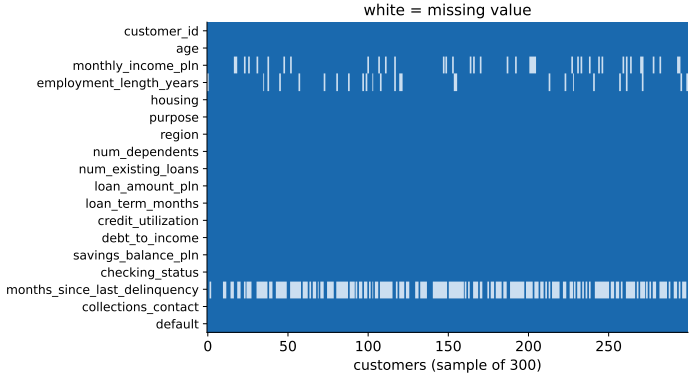
The mechanism decides whether it is safe to drop, impute, or *model*.

Our portfolio: how much is missing, and why



- Three columns, three different mechanisms - and three different correct treatments.
- First step is always the same: count and name the missingness before touching it.

Seeing the pattern of missingness



- A missingness matrix shows *structure*: which columns, how often, and whether gaps co-occur in the same rows.
- In the lab: the same view via `df.isna()` - no extra library needed.

Option 1: deletion - when dropping is (not) safe

Dropping rows (listwise deletion)

- Safe-ish only under **MCAR** - you lose power, but no bias.
- Under MAR/MNAR it biases the sample: dropping income-missing rows here removes high earners.

Dropping columns

- Defensible when a column is mostly empty *and* carries no signal - check both.
- Here, `months_since_last_delinquency` is $\sim$64% missing and still one of the most informative columns in the data.

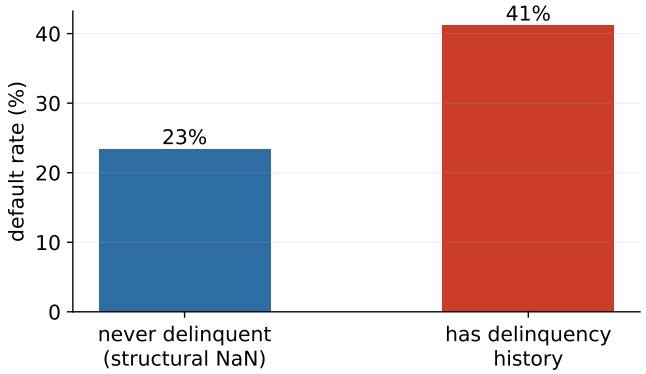
Deletion is a modelling decision, not housekeeping - justify it like one.

Option 2: the imputation menu

Method	Good	Risky
Mean / median	simple, fast; median survives skew	shrinks variance; distorts under MAR/MNAR
Mode (categorical)	simple	inflates the majority level
Group-wise median	respects segments (e.g. per region)	groups must be big enough
Model-based / KNN	uses relationships between features	complexity, leakage risk if fit on all data
Constant + **indicator**	keeps "was missing" as signal	adds columns

Golden rule: fit any imputation on the training data only - a preview of Block 3's leakage discussion.

A missing value can be a signal



- Customers with the structural NaN (never delinquent) default far less often - the gap itself predicts.
- Any treatment that erases the gap erases the signal.

The punchline of Block 1

A missing value can be the strongest signal you have

Customers with *no prior delinquency* carry a structural NaN in "months since last delinquency". Mean-imputing it would invent a delinquency history that never happened - and erase a real signal.

- Honest first move: add a **missing-indicator**, *then* impute the number.
- In regulated lending, the reason you give a customer must be truthful. Imputation choices are not cosmetic - they change the story.

A first warning about leakage

Leakage = using information that will not exist at prediction time

Our dataset contains `collections_contact` - whether collections called the customer. It is recorded after default happens. A model using it looks brilliant in the notebook and is worthless in production.

The time-travel test (ask for every feature):

"Would I have known this value at the moment of the decision?"

- If the answer is no - the feature is a time traveller. Drop it.
- Full treatment (and a live demonstration) in Block 3.

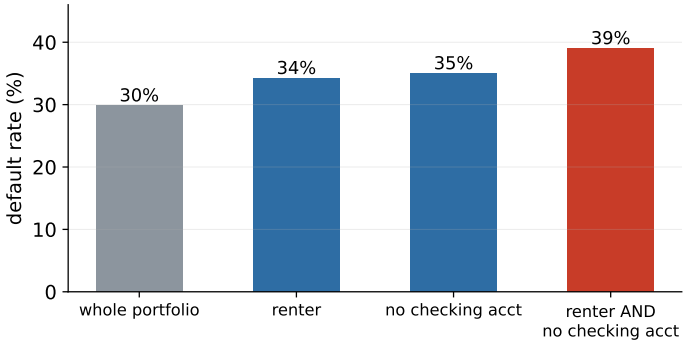
Feature engineering: where domain knowledge becomes signal

New columns that expose structure the model would struggle to find alone. The missing-indicator you just built *is* feature engineering - here is the wider toolbox:

Move	Recipe	Portfolio example
Ratio	divide two raw columns	instalment / income
Flag	boolean condition	`has_delinquency_history`
Binning	continuous → bands	utilisation: low / mid / maxed-out
Interaction	combine two features	renter *and* no checking account
Date arithmetic	differences, tenures	months since last delinquency

Look at the data dictionary again: `debt_to_income` is a ratio - someone engineered it before us.

Worked example: combining two weak flags



- Renting alone and no-checking-account alone are each mildly risky; the interaction flag isolates the riskiest segment (39%, $n = 318$ - large enough to trust).
- And recall the strongest engineered feature so far: the `delinq_missing` indicator ($|r| \approx 0.19$ with default - more than any raw numeric column).

Feature engineering: rules of the game

1. **Application-time information only** - every engineered feature must pass the time-travel test, same as raw ones.
2. **Fit transforms on training data only** - means for scaling, bin edges, encodings (the leakage-safe pipeline, Block 3).
3. **Simple and explainable beats clever** - in regulated lending every feature may end up in an adverse-action reason.
4. **Document the recipe** - production must compute the feature *identically*, or the model silently degrades.
5. **Expect diminishing returns** - five features built from domain sense beat five hundred generated blindly.

Hands-on: exercise 5 in today's lab asks you to engineer one feature from loan amount, term, and income - and defend it.

Encodings: making categories digestible (preview)

Most models eat numbers, not labels. The scale of measurement picks the encoding:

Encoding	Use for	Watch out for
One-hot	nominal (purpose, region)	column explosion with many levels
Ordinal codes	truly ordered (checking status)	only when the order is real
Target encoding	high-cardinality nominal	leakage - fit on train only
WoE (credit tradition)	scorecard features	needs binning first; Block 2

- Encoding nominal as integers invents an order; one-hot on ordinal discards one. *Scale first, encoding second.*
- Full treatment next week, when regression needs numeric inputs.

Reproducibility from day one

- **Fix your random seeds** - `random_state=42` everywhere a coin is flipped (splits, models, samples).
- **Pin your environment** - `uv` lockfile; "works on my machine" is not a result.
- **One-command rule** - a colleague should reproduce your numbers by running one thing, top to bottom.
- **Data provenance** - record where the data came from and when; datasets change under your feet.

If it cannot be reproduced, it is an anecdote, not an analysis - and the project rubric scores it.

Notebooks: superpower and trap

Superpower

- code + plots + narrative in one place
- perfect for EDA and for the project report

Trap: hidden state

- cells run out of order $\Rightarrow$ results that cannot be recreated
- deleted cells leave live variables behind

The habit that saves you

Before trusting any result: **Restart kernel & run all**. If it breaks, it was already broken - you just didn't know.

Version control - yes, for analysts too

- **Commit small and often** - each commit one logical step; messages that say *why*.
- **What goes in**: code, notebooks, configs, figure scripts.
- **What stays out**: raw data (use loaders), credentials, `.env` files, 100 MB artefacts.
- **Why bother in a student project?** You will want yesterday's working version at 23:50 before the deadline.

`git init` today costs 10 seconds. Reconstructing a deleted analysis costs a weekend.

Reading errors & asking good questions

When code explodes

- Read the traceback bottom-up: the last line is the error, the lines above are where it happened.
- Reproduce it in the smallest possible example - half the time the bug evaporates as you shrink it.

When you ask for help (colleague, forum, AI assistant)

- say what you expected, what happened instead, and what you tried
- include the minimal example - not a screenshot of 400 lines
- the discipline of writing the question well often *answers* it

Communicating results to humans

- **Lead with the decision**, not the method: "we can cut losses 12% at the same approval rate" beats "our AUC is 0.79".
- **State uncertainty honestly** - ranges and caveats build trust; false precision destroys it.
- **One chart, one message** - if a slide needs a paragraph of explanation, it is two slides.
- **Know your audience** - the risk committee, the regulator, and your fellow data scientist need three different stories.

Your model's impact is capped by your ability to explain it.

Beginner traps (we will meet them all)

1. **Evaluating on training data** - your score is a fantasy (Block 2).
2. **Metric worship** - optimising accuracy on a 97/3 problem (Block 3).
3. **Complexity first** - neural network before a logistic baseline; you can't tell if the extra complexity earned anything (Blocks 2, 6).
4. **Not looking at the data** - today's whole point.
5. **Torturing the data until it confesses** - test 100 hypotheses, report the 5 that "worked".
6. **Silently deleting inconvenient rows** - outliers and missing values carry signal.

The course subtitle is a promise: how not to fool yourself.

Laboratory

Notebook `01_eda_missingness` - laptops out.

In the lab today

1. **Load** the portfolio and run the first-contact commands.
2. **Profile** the target and the key numeric / categorical features (the plots from today, hands-on).
3. **Hunt** the missingness: count it, plot it, name the mechanism for each column.
4. **Experiment**: mean-impute vs indicator-plus-impute - watch what happens to the delinquency signal.

- Runs locally or on Colab; data loads itself. Light on Python? Start with `00_python_primer`.

"Done" = you can defend, in one paragraph, how you would treat each missing column and why.

Summary

What to take home from Block 1.

Block 1 in five sentences

1. Data mining starts from a **decision**, and translating it into a learning task is your job, not the algorithm's.
2. We focus on **binary classification** and **regression**, with the rest of the map mentioned as we pass it.
3. **EDA before models**: target balance, distributions, bivariate relationships, and always plot (Anscombe!).
4. Missingness has a **mechanism** (MCAR / MAR / MNAR / structural) - and the mechanism dictates the treatment.
5. A missing value can be **signal**: indicator first, impute second, and never invent a history that didn't happen.

Project reminder

- Before next class (Block 2, 13.10): teams of 3, formed and registered. That is this week's only project task.
- Looking one week further: topic + short description due **before Block 3** - if no idea comes, consult early.

Full brief, timeline, grading and the AI rule: the project slides at the start of today's deck.

Exam practice - multiple choice

Q1. `months_since_last_delinquency` is NaN exactly for customers who were *never* delinquent. The missingness mechanism is:

a. MCAR - missing completely at random
b. MAR - explained by another observed column
c. MNAR - depends on the unobserved value itself
d. structural - the value cannot exist

Q2. A default model reports 92% accuracy on a portfolio with an 8% default rate. The honest reading:

a. excellent - deploy it
b. says little: predicting "repaid" for everyone also scores 92%
c. the model is overfit
d. accuracy is too low for credit work

Exam practice - open question

In the style of the term-0 exam (∼60% open questions)

A bank wants a default model for *all future applicants*, but its dataset contains only loans it previously *approved*. Explain the problem this creates for the model, name the phenomenon, and propose one mitigation.

Write 4-6 sentences. Naming the concept is a third of the credit; explaining the consequence is the rest.

Next week: regression, re-framed.

From *inference* to *prediction*.

Keep learning - Block 1 reading

- **Shearer** - *The CRISP-DM Model: The New Blueprint for Data Mining* (J. of Data Warehousing, 2000) - the process spine of this course, from the source.
- **Anscombe** - *Graphs in Statistical Analysis* (The American Statistician, 1973) - today's quartet, in the author's own words.
- **Rubin** - *Inference and Missing Data* (Biometrika, 1976) - where MCAR / MAR / MNAR actually come from.
- **Wickham** - *Tidy Data* (J. of Statistical Software, 2014) - how to shape a table before any analysis touches it.

Course-wide references (ISLR, Géron, the scikit-learn User Guide) live in the repo README.